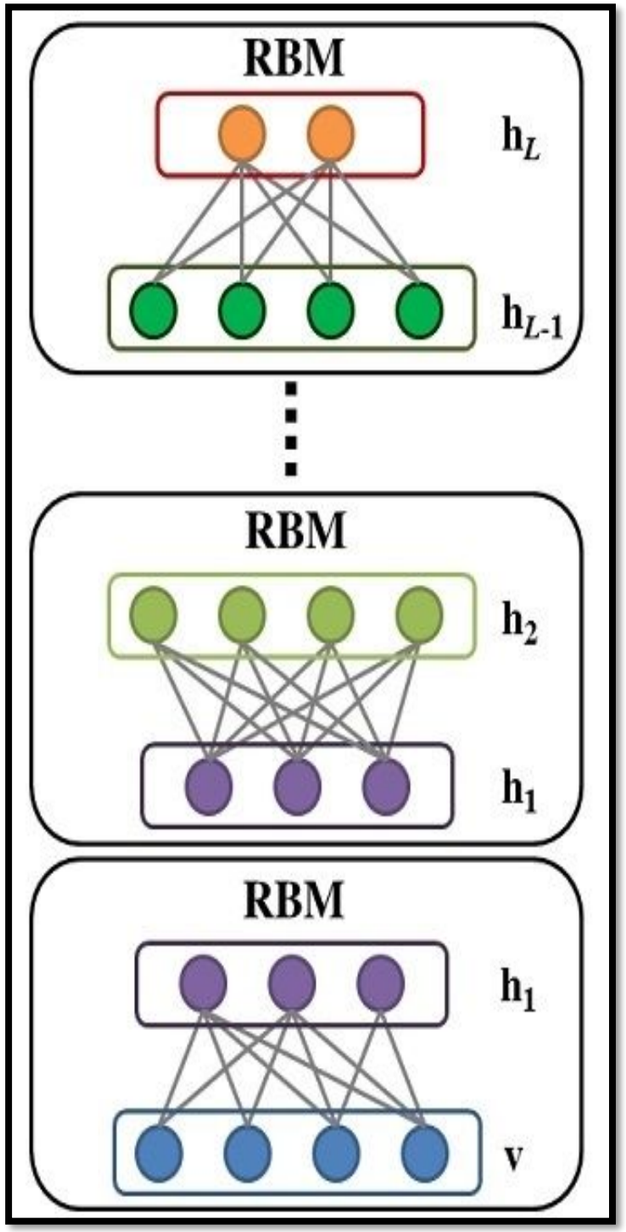
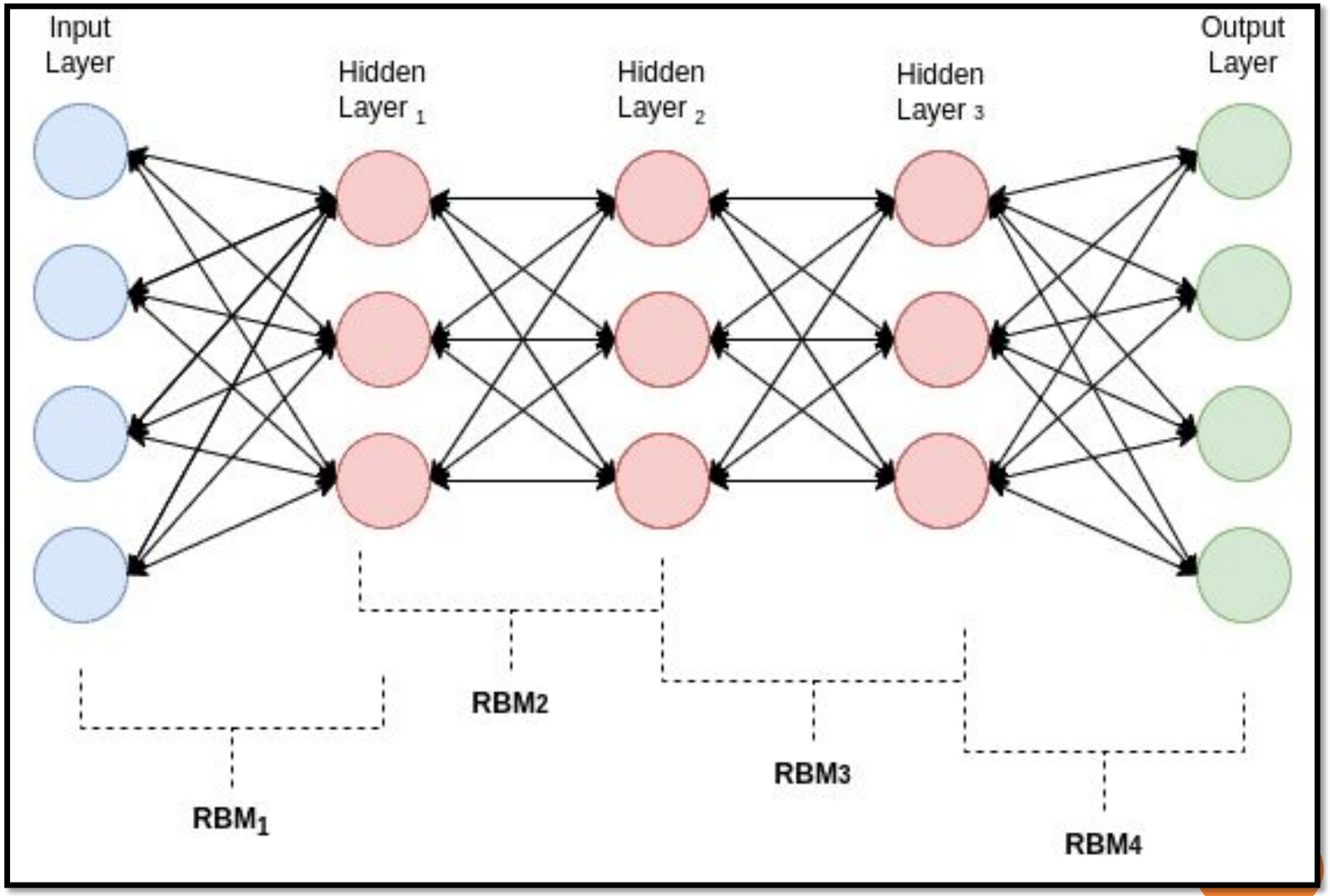
UNIT-II

Deep Belief Networks

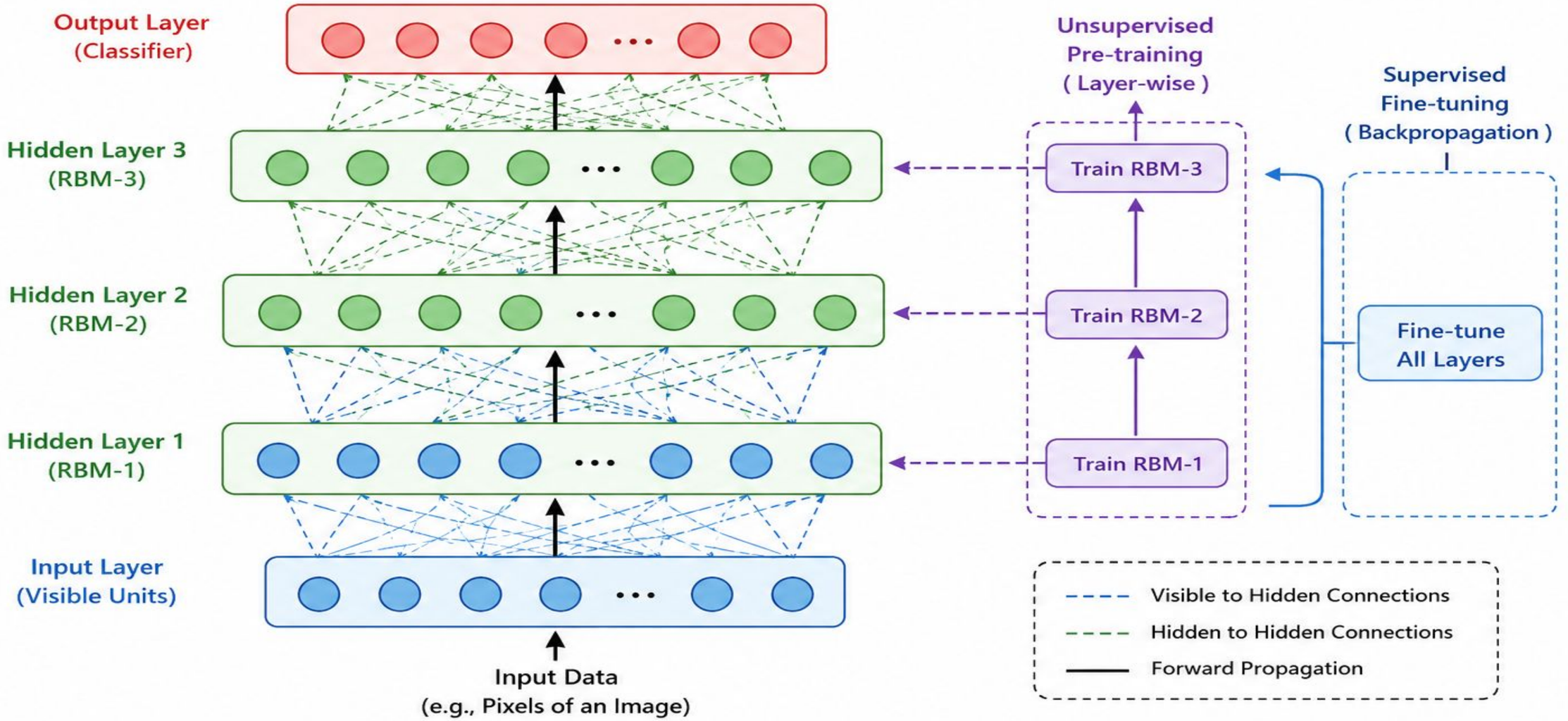
INTRODUCTION

- We create ***Deep Belief Networks (DBNs)*** to **address issues** with **classic neural networks** in **deep layered networks.**
- **For example – slow learning, becoming stuck in local minima problem because of poor parameter selection, and requiring a large number of training datasets for the given input layer.**
- It was introduced by **Geoffrey Hinton in 2006. Deep Belief Networks (DBNs)** are a type of **deep learning architecture** combining **unsupervised learning principles and neural networks.**
- **A Deep Belief Network (DBN)** is a type of **deep generative neural network** made by **stacking multiple Restricted Boltzmann Machines (RBMs).**
- DBNs were among the **first deep learning models that successfully trained very deep**


RBM
h_L
h_{L-1}
RBM
h_2
h_1
RBM
h_1
v


Input Layer
Hidden Layer 1
Hidden Layer 2
Hidden Layer 3
Output Layer
RBM1
RBM2
RBM3
RBM4

Deep Belief Network (DBN) Architecture

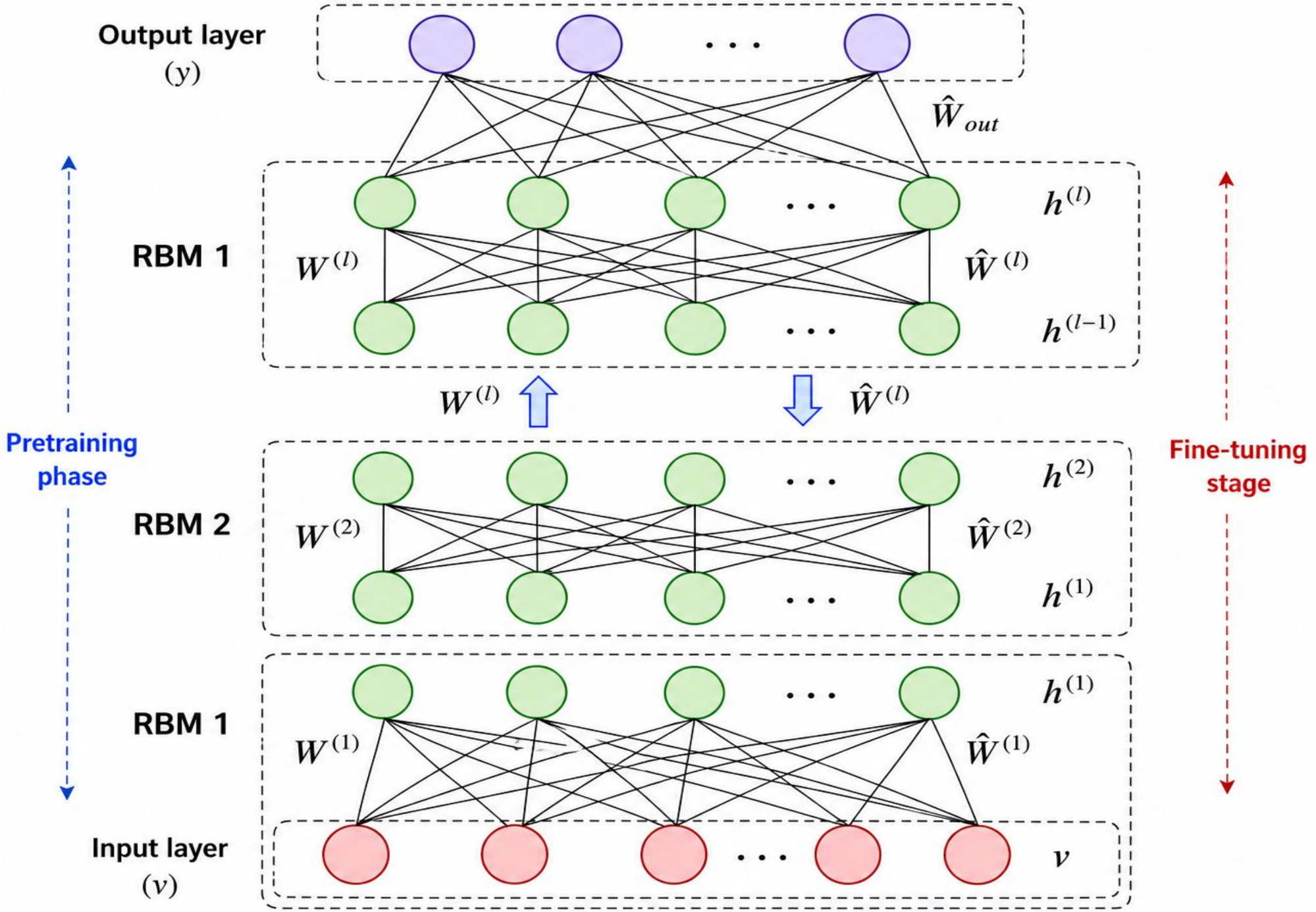


DBN = Stack of RBMs + Classifier
Learns hierarchical feature representations from input (low-level to high-level).

- They are designed to **discover and learn patterns within large sets** of data automatically.
- Imagine them as **multi-layered networks**, where **each layer is capable of making sense of the information received from the previous one,** gradually **building up a complex understanding of the overall data.**
- They are **composed of layers** of **Restricted Boltzmann Machines (RBMs),** which are **trained one at a time in an unsupervised manner.**
- **Several Restricted Boltzmann Machines (RBM)** can be **stacked** and **trained in a greedy manner** to **form Deep Belief Network architecture.**
- i.e, It is a **composition of Stack** of **Unsupervised Neural Network** such as **Restricted Bolzmann Machine(RBM). Each RBM** has a **Visible Layer (Input)** and **Hidden Layer (Output)**

- Here the **Hidden Layer in Stack1** is the **Visible Layer for the Stack2.etc.**
- The **output of one RBM** is used as the **input to the next RBM,** and the **final output** is used for **supervised learning tasks** such as **classification or regression**.
- Here, **Each RBM network is trained independently** with their **greedy method.**
- DBNs work similarly to **traditional multi-layer perceptron's (MLPs)** and offer certain benefits over them, including **faster training and better weight initialization.**




Output layer
(y)
$\hat{W}_{out}$
RBM 1
$W^{(l)}$
$\hat{W}^{(l)}$
$h^{(l)}$
$h^{(l-1)}$
$W^{(l)}$
$\hat{W}^{(l)}$
Pretraining phase
Fine-tuning stage
RBM 2
$W^{(2)}$
$\hat{W}^{(2)}$
$h^{(2)}$
$h^{(1)}$
RBM 1
$W^{(1)}$
$\hat{W}^{(1)}$
$h^{(1)}$
Input layer
(v)
v

- In the DBN, we have a **hierarchy of layers**.
- "The **top two layers are associative memory**" means that the **highest two layers of a DBN** learn and **represent associations among high-level features**, allowing the network to **capture and generate meaningful patterns from the learned data.**, and the **bottom layer is the visible units**.
- The **arrows pointing** towards the **layer closest to the data point** to **relationships between all lower layers**.



How Deep Belief Networks Works?

- The Deep Belief Network (DBN) algorithm consists **of two main steps:**
  - **1. Pre Training**
  - **2. Fine Tuning**

1. Pre Training: In the **pre-training phase**, the **network learns to represent the input data layer by layer**. **Each layer** is **trained independently as an RBM**, which allows the **network to learn complex data representations efficiently.**

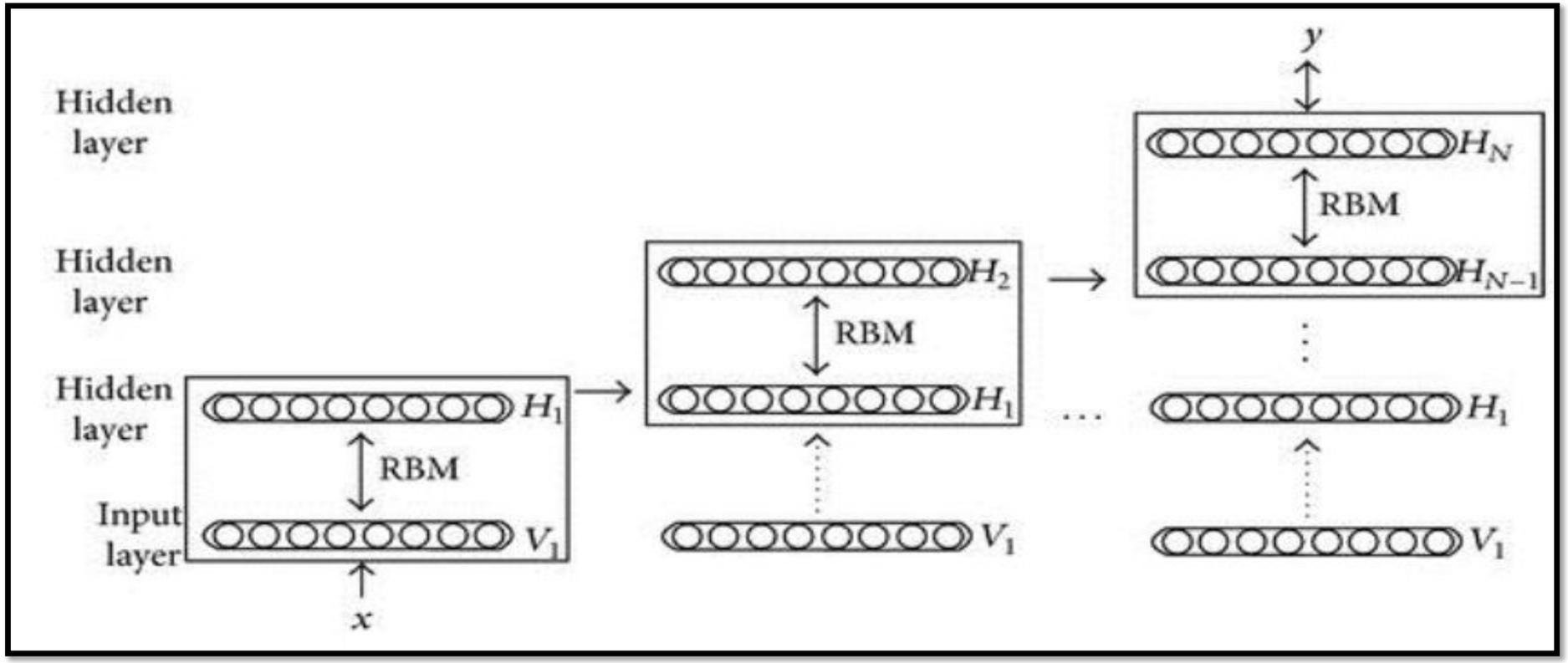
- The **ensemble's first layer (often called the input layer or bottom layer) interacts directly with the raw data, learning its features** and **creating a latent representation during the process.**

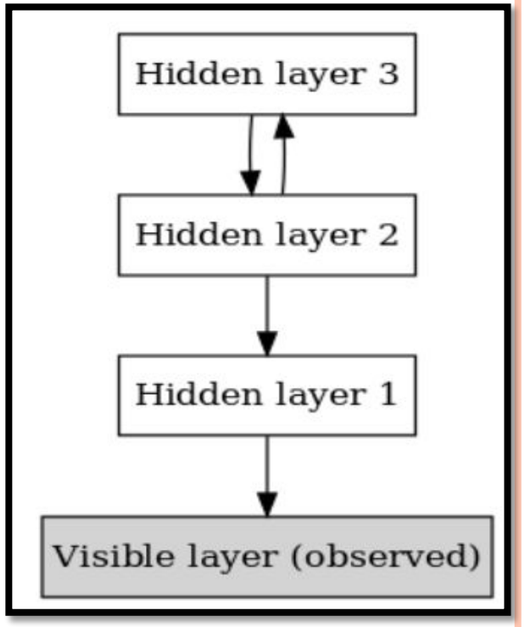
- Each **subsequent layer** is **then trained so the first's output** becomes **the next's input**.
- This **greedy layer-wise learning** allows for **efficient feature learning.**
- We **iterate this process multiple times, covering various data samples** and **updating weights after each pass.** Finally, the **last layer (output layer) outputs** the **network's prediction**.



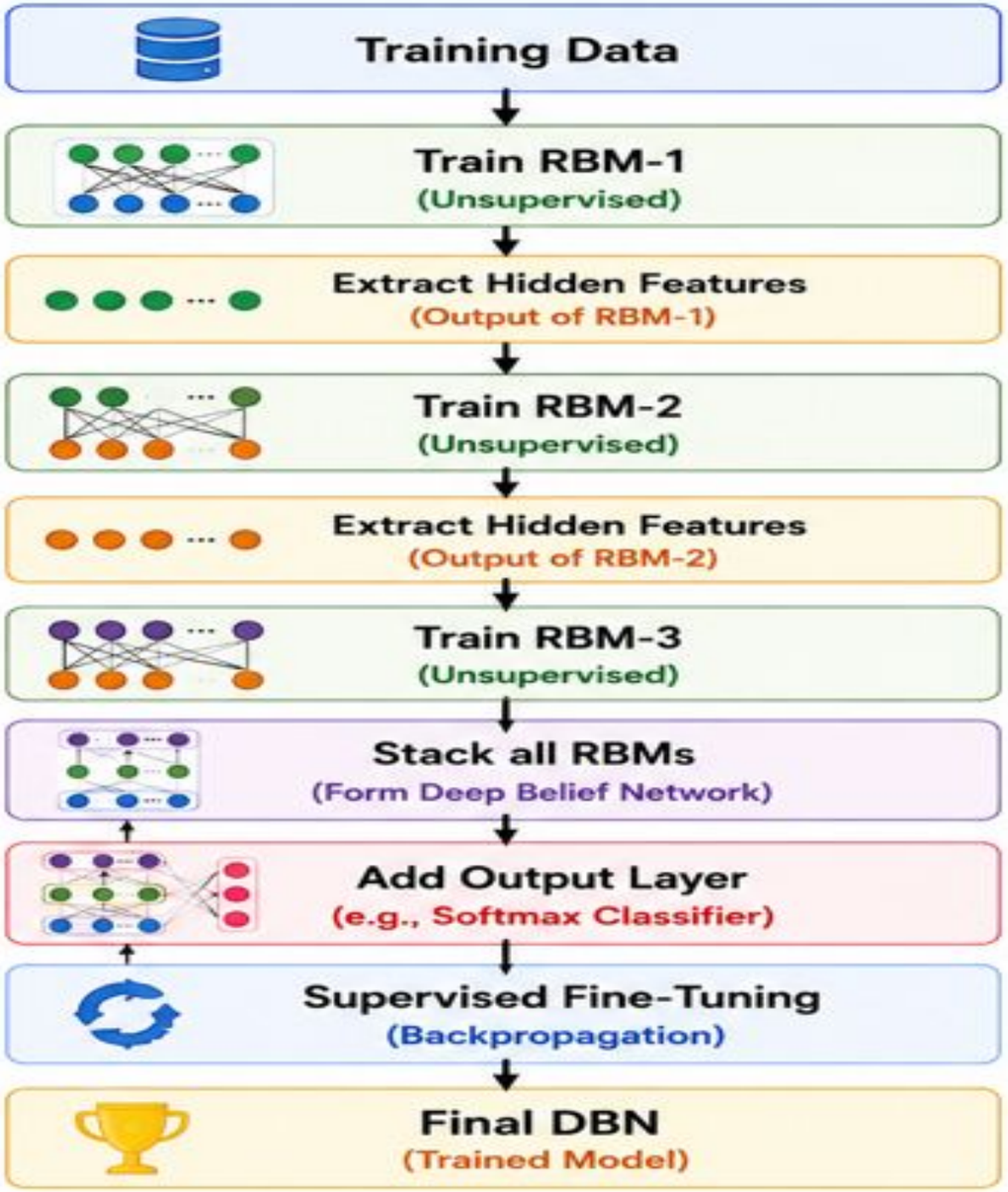
2. Fine Tuning: In the **fine-tuning phase**, the **DBN adjusts its parameters for a specific task,** like **classification or regression**.

- This is typically done using a **technique known as backpropagation**, where the **network's performance on a task is evaluated**, and the **errors are used to update the network's parameters**.
- This phase often involves **supervised learning**, where the **network is trained with labelled data.**

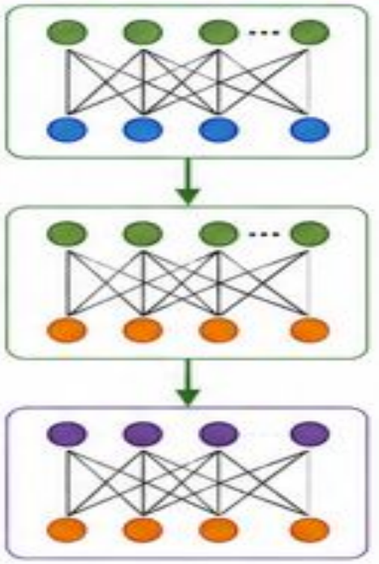




DBN TRAINING PROCEDURE



STAGE 1: UNSUPERVISED PRE-TRAINING (LAYER-WISE)



RBM-1

Learns simple features
(e.g., edges, lines, corners)
from raw input data.

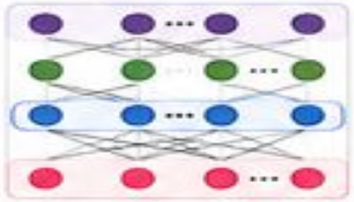
RBM-2

Learns higher-level features
(e.g., shapes, parts)
from features of RBM-1.

RBM-3

Learns more abstract features
(e.g., objects, patterns)
from features of RBM-2.

STAGE 2: SUPERVISED FINE-TUNING



All layers (RBMs + Output layer)
are fine-tuned together using
backpropagation with labeled data
to improve prediction performance.

LEGEND

- Hidden Units of RBM-1
- Hidden Units of RBM-2
- Hidden Units of RBM-3
- Visible Units (Input)
- Output Units (Classes)

KEY TAKEAWAY

DBN is trained in two phases:

- **Unsupervised Pre-training**
  (layer-wise using RBMs)
- **Supervised Fine-Tuning**
  (backpropagation)

This enables DBN to learn rich,
hierarchical feature representations.

Mathematical Representation

- **Deep Belief Networks (DBNs)** employ **several mathematical concepts, blending probability theory** with **neural network structures.** At their core, **they use Restricted Boltzmann Machines (RBMs)** for **layer-wise learning**, which are **based on probabilistic graphical models.**
- **1. Energy-Based Model:** **Each RBM within a DBN is an energy-based model.** For an RBM with **visible units v** and **hidden units h**, the **energy function** is defined as: $E(v,h) = -\sum_i a_i v_i - \sum_j b_j h_j - \sum_{i,j} v_j h_j w_{ij}$
- Here, **ai and bj** are **bias terms**, and **wij** represents the **weights between units**.

2. Probability Distribution: The **probability of a given state** of the **RBM** is defined by the **Boltzmann distribution:**

$$P(v,h) = \frac{e^{-E(v,h)}}{Z}$$

where **Z is the partition function**, a **normalization factor** calculated as the **sum over all possible pairs of visible and hidden units**.

Step 1 – Input Image

Handwritten digit



Pixel values (5×5 example)

0	1	1	1	0
1	0	0	0	1
0	1	1	1	0
1	0	0	0	1
0	1	1	1	0

(Visible layer of DBN)

Step 2 – RBM-1 Learns Basic Features

Pixels

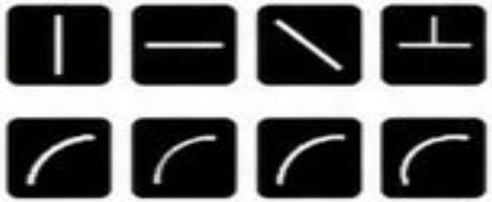
↓

RBM-1

↓

Edges / Lines

Examples of basic features learned



- Vertical lines
- Horizontal lines
- Curves
- Edges

Step 3 – RBM-2 Learns Shapes

Edges (from RBM-1)

↓

RBM-2

↓

Shapes

Examples of shapes learned



Top loop | Bottom loop | Circular shape

Step 4 – RBM-3 Learns High-Level Feature

Shapes (from RBM-2)

↓

RBM-3

↓

High-level pattern

Combination of shapes



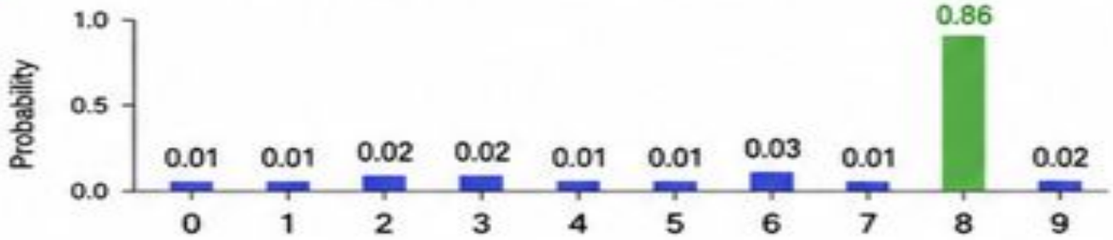
Two connected loops

→ Possible Digit = 8

Step 5 – Classification Layer (Softmax)

High-level features (from RBM-3) → Softmax Classifier →

Output probabilities for digits 0–9



Predicted Digit = **8**

Benefits of DBN

- Deep belief networks use **probabilistic modeling** to offer **certain benefits** over **conventional neural networks.** **These include:**
- **Ability to handle large data using hidden units** to **extract underlying correlations.**
- **Faster training and better results**. **Achieving global minima due to better weights initialization.**
- **Model Interpretability:** Like many deep learning models, DBN is called a **black-box model** because we **can observe its input and prediction,** but it **can be difficult to clearly explain the internal features** and **computations responsible for that prediction.**

Applications of DBN

Deep Belief Networks (DBNs) have been **applied** in a **variety of fields, including:**

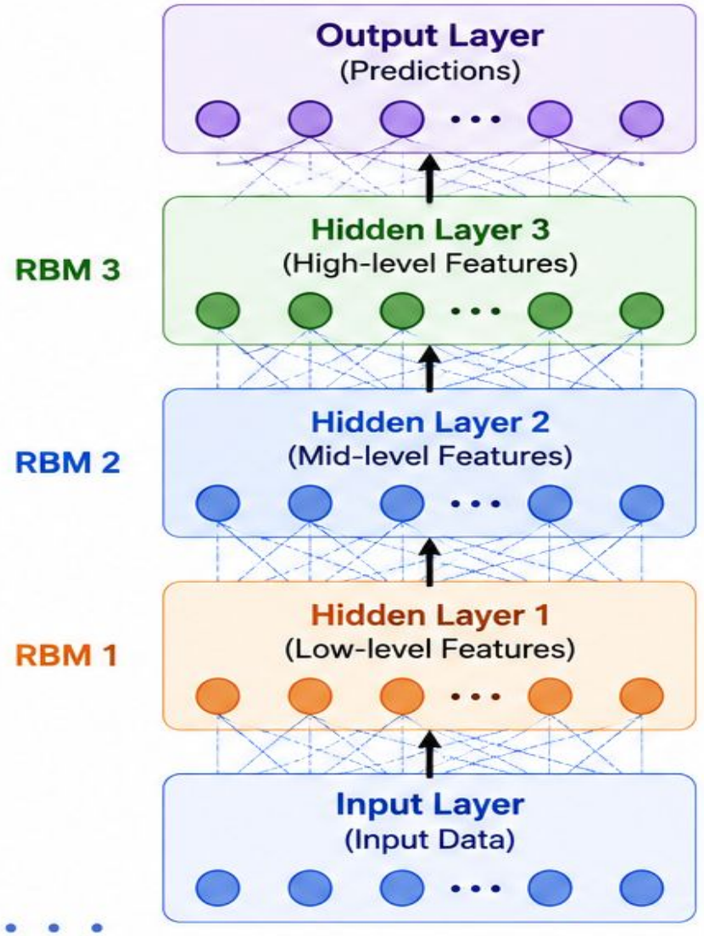
- **Computer vision**: DBNs have been used for tasks like **recognizing objects in pictures** or **putting pictures into different groups.**
- **Speech recognition**: DBNs have been used for **speech recognition tasks,** such as **transcribing speech into text.**
- **Natural language processing**: DBNs have been used for natural languages processing tasks, such as sentiment analysis and text classification.
- **Recommender systems:** DBNs have been used in recommender systems, which give users suggestions based on what they like and how they act.
- **In bioinformatics,** DBNs have been used to **predict how proteins will interact with each other,** look at **how genes are expressed**, **and find new drugs.**
- **Financial analysis:** DBNs have been used to **predict the stock market** and **figure out how risky something is**.

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Using Deep Belief Network (DBN)



Thank You!

For Your Attention

Input Data
Raw data is provided to the network

Layer-wise Pre-training
RBMs are trained one by one in an unsupervised manner

Feature Learning
Hierarchical features are learned from low-level to high-level

Fine-tuning
All layers are fine-tuned using labelled data

Prediction
The trained DBN makes accurate predictions